

SOFTWARE ARCHITECTURE DESCRIPTION

CSI-100 Software System (IVD)

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Purpose

This document describes the architecture of the CSI-100 software system: software items and units, interfaces, data flows, safety classification and segregation, SOUP, design decisions and the integration strategy, per IEC 62304 clause 5.3. It implements SRS-SW-CSI-010 Rev E.

Document Control

Rev	Date	Software version	Description of change	Author
A	Mar 2025	1.0	Initial architecture: workstation application and instrument firmware.	A. Lodhi
B	Mar 2026	2.5	Added records service, audit trail and e-signature service, security service; safety supervisor MCU added to firmware; cybersecurity architecture.	A. Lodhi
C	Aug 2026	3.0	Split workstation software into Research, Admin and Data Management applications; added board management FPGA, fail-safe state model, internal log and protocol architecture; integration strategy updated; allocation aligned with SRS-SW-CSI-010 Rev E.	A. Lodhi

Approvals

Function	Role	Signature / Date
Software Engineering	Software Architect (author of Parts A, C, D)	
Firmware Engineering	Firmware and FPGA Architect (Part B)	
Systems Engineering	System Architect	
Software Quality Assurance	Software Quality	
Quality Engineering	Risk Management	
Cybersecurity	Product Security Lead	

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1. Introduction

1.1 Purpose

This Software Architecture Description (SAD) defines the architecture of the CSI-100 software system. It meets IEC 62304 clause 5.3: it transforms the software requirements into an architecture of software items (5.3.1), defines their interfaces (5.3.2), specifies the functional and performance requirements and the hardware and software needed by SOUP (5.3.3, 5.3.4), identifies the segregation needed for risk control (5.3.5), and records the verification of the architecture (5.3.6). It also defines the integration strategy used under clause 5.6.

The SAD is the technical backbone of the software design file. The detailed design of each software unit is in SDS-SW-CSI-012 (workstation) and SDS-FW-CSI-013 (firmware and FPGA).

1.2 Scope

This document covers software release 3.0 and all software items defined in SRS-SW-CSI-010 Rev E: the Research Application (Part A), Firmware (Part B), Admin Application (Part C), Data Management Application (Part D) and the shared services, for all laser and detector configurations in CFG-CSI-001.

1.3 Architectural views

View	Section	Question it answers
Context	3	What is outside the software system and how does it interact with it?
Deployment	4	On which processors and processes does each software item run?
Decomposition	5, 6, 7	Which software items and units exist, and what is each responsible for?
Interface	8	How do items communicate, with what data, rate and safety relevance?
Dynamic	9	How do data and control flow in the scenarios that matter most?
Safety	10	Which items are safety-related, how are they segregated, and what is the safe state?
Data and security	11, 12	Where is data stored and how is it protected?

1.4 Conventions

Software items are identified SI-nn and software units SU-nn.m, where nn is the parent item. Requirement identifiers (RA, FW, AD, DM, CM) are those in SRS-SW-CSI-010. Design decisions are identified ADR-nn.

2. Architectural Drivers and Principles

2.1 Key drivers

Driver	Source requirements	Architectural consequence
Deterministic sort decisions in under 20 μ s at 70,000+ events/s	FW-202, FW-204	Sort path implemented in FPGA logic, not software on a CPU (ADR-01)
Safe response to clog, e-stop and link loss without the workstation	FW-601 to FW-605	Autonomous fault manager in firmware; independent safety MCU (ADR-02, ADR-03)
Different users, permissions and change control for routine use, calibration and records	Section 3.4 of SRS-SW	Three separate applications on shared services (ADR-04)
Data integrity and 21 CFR Part 11	DM-103, DM-201 to DM-304	Single records service owns the database and the audit trail (ADR-05)
Configuration-dependent optics across 16 builds	AD-101 to AD-106	Configuration-driven software: optical configuration generated from a controlled record (ADR-07)
Robust instrument link	FW-801 to FW-808	Channelized TCP protocol with sequence numbers, CRC and heartbeats (ADR-06)
Cybersecurity	CM-201 to CM-203, FW-104	Locked workstation image, paired instrument, signed updates (ADR-09)

2.2 Design principles

- **Safety does not depend on the workstation.** Every safety-related response runs in firmware or hardware.
- **Hardware first for serious hazards.** Laser, high voltage and pressure hazards are controlled by hardware; software adds secondary controls.
- **One owner for each piece of data.** The records service is the only writer to the database; firmware is the only writer to the internal log.
- **Separation by role.** Routine operation, calibration and record management are separate applications with separate permissions.
- **Configuration over code.** Optical builds, sort setups and precision modes are controlled data, not compiled constants.
- **Testable in isolation.** Every item has a defined interface that can be simulated for unit and integration testing.

3. System Context

Figure 1 shows the software system and the people and systems it interacts with. The instrument hardware is outside the software boundary; it is controlled through the firmware.

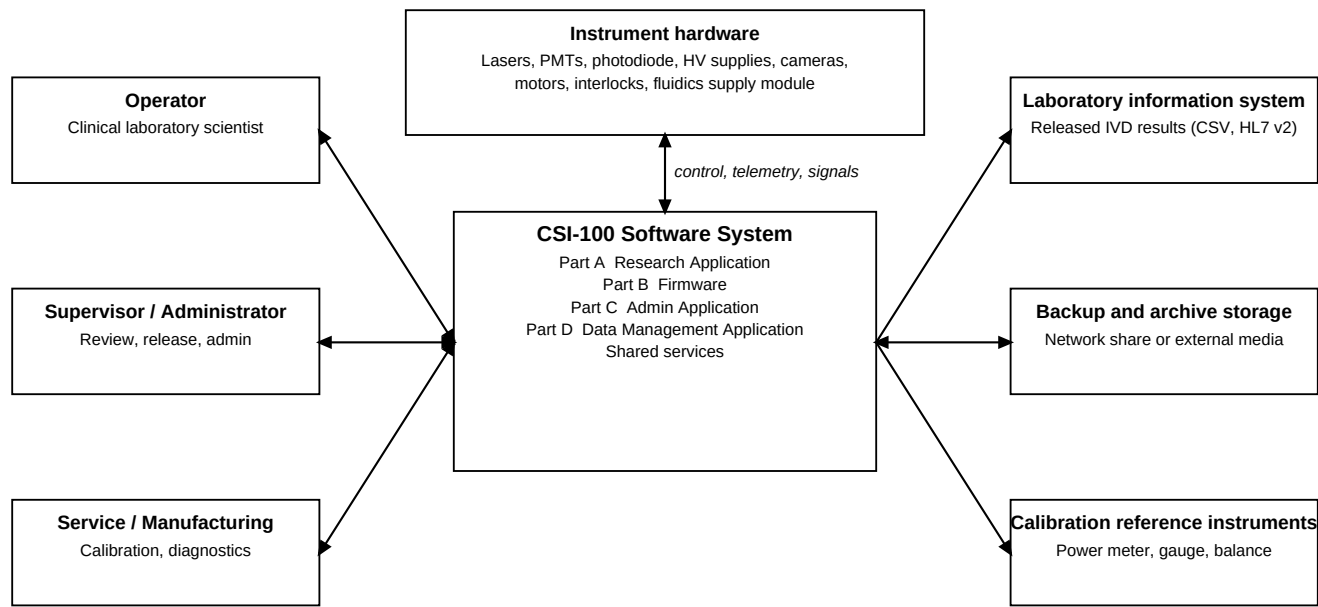


Figure 1. System context.

External entity	Interaction	Interface
Operator	Runs startup, QC, acquisition, sorting and shutdown in the Research Application	IF-05 (user interface)
Supervisor / administrator	Reviews and releases results, approves assays, administers users, backup and archive	IF-05
Service engineer / manufacturing	Configures hardware, calibrates, reads diagnostics and internal log in the Admin Application	IF-05, IF-07
Instrument hardware	Controlled and monitored by firmware	IF-03, IF-04, hardware signals
Laboratory information system	Receives released IVD results	IF-06
Backup and archive storage	Receives backups and archives; returns restored data	IF-06
Calibration reference instruments	Provide reference readings for calibration	IF-07

4. Deployment View

The software runs on three hardware nodes (Figure 2). The workstation runs three application processes and one Windows service. The sorter contains the controller system-on-chip (processor cores running the real-time operating system, plus programmable logic for the sort path), board-level FPGAs, an imaging processor and an independent safety microcontroller. The fluidics supply module has its own microcontroller.

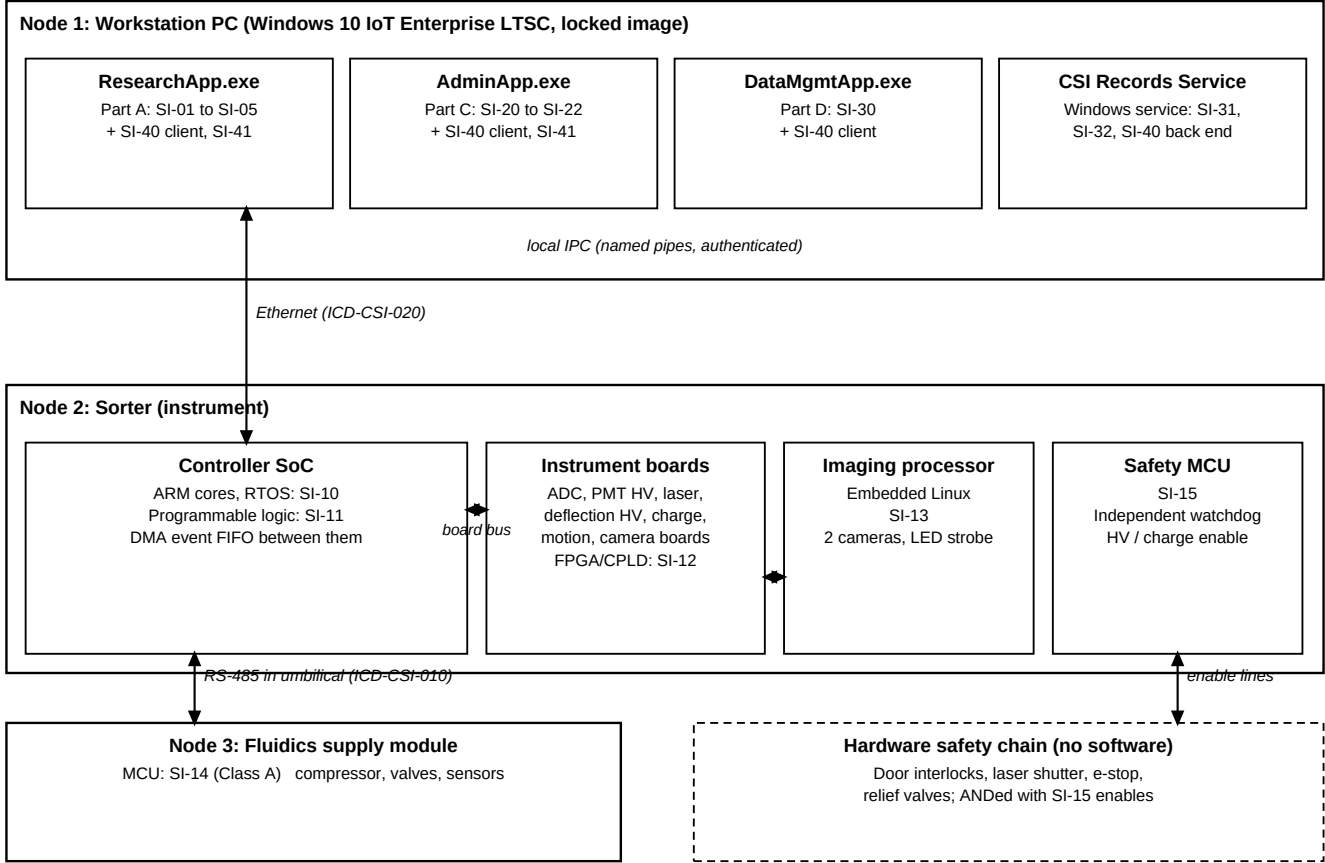


Figure 2. Deployment view: nodes, processors, processes and links.

Node / processor	Software	Execution environment
Workstation PC	ResearchApp.exe, AdminApp.exe, DataMgmtApp.exe, CSI Records Service	Windows 10 IoT Enterprise LTSC; .NET LTS; locked image
Controller SoC, processor cores	SI-10	Certified RTOS, fixed-priority preemptive scheduling
Controller SoC, programmable logic	SI-11	FPGA fabric, 200 MHz pipeline clock
Instrument boards	SI-12 (one image per board type)	FPGA or CPLD per board
Imaging processor	SI-13	Embedded Linux, real-time priority for image analysis
Safety MCU	SI-15	Bare-metal, independent clock and watchdog
Fluidics module MCU	SI-14	Bare-metal

Only one application may hold the instrument control session at a time. The Research Application holds it during operation; the Admin Application requests an exclusive service session, which firmware grants only when no sort or

acquisition is running. The Data Management Application never connects to the instrument.

5. Software Items and Safety Classification

The software system is Class B (rationale in SRS-SW-CSI-010 Section 2). Table 1 lists every software item, its part, safety class, the process or processor it runs in, and the number of software requirements allocated to it in SRS-SW-CSI-010.

Item	Name	Part	Class	Runs in	Reqs
SI-01	Operator UI and Workflow	A	B	ResearchApp.exe	13
SI-02	Acquisition and Analysis	A	B	ResearchApp.exe	10
SI-03	Sort Control	A	B	ResearchApp.exe	22
SI-04	Fluidics and Instrument Control	A	B	ResearchApp.exe	9
SI-05	QC Run and Assay Execution	A	B	ResearchApp.exe	14
SI-10	Instrument Controller Firmware	B	B	Controller SoC cores	36
SI-11	Sort Logic FPGA	B	B	Controller SoC logic	10
SI-12	Board Management FPGA	B	B	Instrument boards	14
SI-13	Stream Imaging Firmware	B	B	Imaging processor	7
SI-14	Fluidics Module Firmware	B	A	Fluidics module MCU	5
SI-15	Safety Supervisor Firmware	B	B	Safety MCU	3
SI-20	Hardware Configuration Control	C	B	AdminApp.exe	8
SI-21	Measurement and Calibration Tools	C	B	AdminApp.exe	26
SI-22	User, Assay and Bead Lot Administration	C	B	AdminApp.exe	8
SI-30	Data Management Application	D	B	DataMgmtApp.exe	16
SI-31	Database and Records Service	D	B	CSI Records Service	9
SI-32	Audit Trail and E-Signature Service	D	B	CSI Records Service	10
SI-40	Security Service	Shared	B	Records Service + client in each app	6
SI-41	Instrument Communication Library	Shared	B	ResearchApp.exe, AdminApp.exe	9

Table 1. Software items. Requirement counts are computed from SRS-SW-CSI-010 Rev E.

Class A item SI-14 is segregated on its own processor (Section 10.2). All other items are Class B and are developed with the Class B activities of IEC 62304, including detailed design and unit verification of every unit.

6. Workstation Architecture (Parts A, C, D and shared services)

6.1 Structure

Each application is a separate Windows process built on the same layered structure (Figure 3): a presentation layer (views, no business logic), an application services layer (workflows and use cases), a domain layer (rules and models), and shared infrastructure libraries. Layers call only downward. The CSI Records Service is a separate Windows service that owns the database, the audit trail and the security back end; applications reach it only through an authenticated local IPC interface.

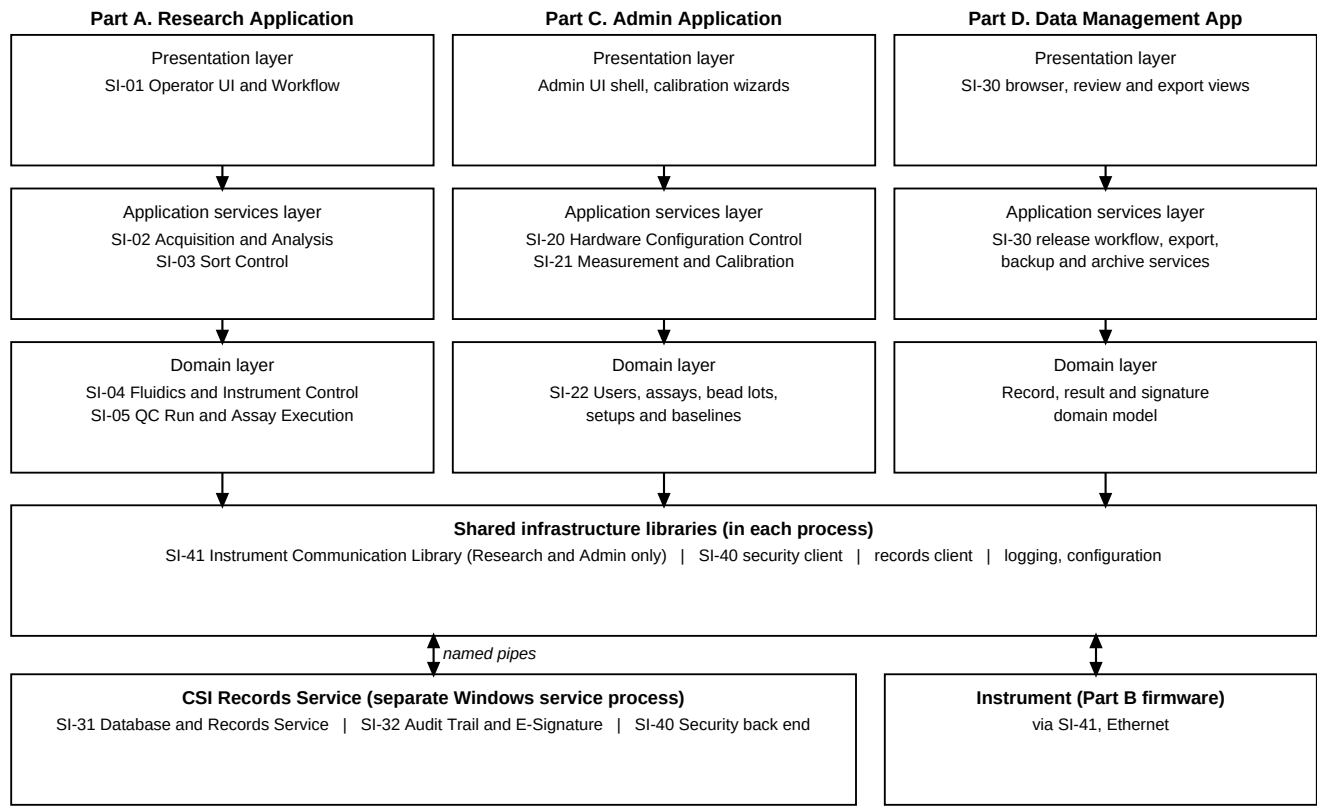


Figure 3. Workstation software structure.

6.2 Part A: Research Application decomposition

Unit	Name	Responsibility
SI-01	Operator UI and Workflow	
SU-01.1	Shell and navigation	Main window, menus, workspace layout
SU-01.2	Plot views	Dot plots, histograms, density plots; rendering from SI-02 data
SU-01.3	Guided procedure engine	Step-by-step procedures for startup, QC, sort, shutdown
SU-01.4	Alarm presenter	Alarm list, priorities, required actions, acknowledgement
SU-01.5	Status panel	Fluids, pressure, lasers, HV, configuration, QC and link status
SI-02	Acquisition and Analysis	
SU-02.1	Acquisition manager	Starts and stops acquisition, detector settings, thresholds
SU-02.2	Event decoder	Decodes event packets from SI-41 into parameter arrays
SU-02.3	Compensation engine	Spillover matrix calculation and application
SU-02.4	Gating and statistics engine	Gate hierarchy evaluation, statistics
SU-02.5	Experiment model	Experiments, tubes, parameters, settings for storage
SI-03	Sort Control	
SU-03.1	Sort setup manager	Loads controlled default and custom setups per nozzle
SU-03.2	Breakoff controller	Closed-loop amplitude adjustment on Drop 1 and Gap
SU-03.3	Drop delay routine	Automatic drop delay sweep and selection
SU-03.4	Layout and device manager	Collection devices, layouts, precision modes, gate tables
SU-03.5	Sort session manager	Start, pause, resume, stop, counters, pause log
SU-03.6	Sort report generator	Sort report content (RA-706)
SI-04	Fluidics and Instrument Control	
SU-04.1	Procedure sequencer	Scripted fluidics startup, shutdown and cleaning
SU-04.2	Sample handling	Load, unload, flow rate, agitation, temperature
SU-04.3	Fluid status monitor	Levels, pressures, capacity check before sort
SU-04.4	Options manager	Aerosol evacuation, temperature control, plate unit
SI-05	QC Run and Assay Execution	
SU-05.1	Performance check runner	Runs QC beads, computes results against baseline
SU-05.2	QC evaluator and lockout	Pass/warn/fail, 24-hour lockout, override record
SU-05.3	Assay template runtime	Executes locked IVD templates; result flags
SU-05.4	Configuration match checker	Compares active lasers with selected configuration

The Research Application holds the instrument control session. SU-03.4 compiles gates and sort layouts into gate lookup tables and mask settings, which are downloaded to firmware before a sort starts; during the sort the application only monitors and records.

6.3 Part C: Admin Application decomposition

Unit	Name	Responsibility
SI-20	Hardware Configuration Control	
SU-20.1	Configuration record editor	Lasers, detectors, optics, fluidics, options; versioning
SU-20.2	Inventory comparator	Compares record with firmware board inventory
SU-20.3	Configuration rules engine	Applies CFG-CSI-001 rules
SU-20.4	Optical configuration generator	Generates parameter and detector maps for Part A
SI-21	Measurement and Calibration Tools	
SU-21.1	Laser calibration	Power, warm-up, inter-laser delay and area scaling
SU-21.2	Optics and detector calibration	Alignment check, PMT gain curves, linearity, resolution
SU-21.3	Fluidics calibration	Pressure sensors, flow rate, level sensors, leak test, temperature
SU-21.4	Sort system and stage calibration	Cameras, drop drive, plate voltage, stream position, plate stage
SU-21.5	Calibration record manager	As-found/as-left, due dates, certificates, DHR format
SU-21.6	Diagnostics and log viewer	Firmware telemetry, board status, internal log export
SI-22	User, Assay and Bead Lot Administration	
SU-22.1	User administration	Accounts and roles through SI-40
SU-22.2	Assay template approval	Versioning and approval signature
SU-22.3	Bead lot manager	Import, integrity check, expiry
SU-22.4	Setup, device and mode administration	Default setups, custom devices, precision modes
SU-22.5	Baseline manager	Performance check baselines per configuration

The Admin Application publishes controlled records (hardware configuration, optical configuration, default setups, baselines, bead lots, calibration records) through the records service. The Research Application reads only released versions of these records.

6.4 Part D: Data Management Application and records service decomposition

Unit	Name	Responsibility
SI-30	Data Management Application	
SU-30.1	Record browser and search	Browse and search all record types
SU-30.2	Review and release workflow	Draft, reviewed, released; versioning of released records
SU-30.3	Export	FCS 3.1, PDF, CSV, LIS (CSV or HL7 v2), de-identification, checksums
SU-30.4	Backup, restore and archive	Scheduled and manual backup; verified restore; archive and retrieval
SU-30.5	Retention manager	Retention periods, soft delete, storage monitoring
SI-31	Database and Records Service	
SU-31.1	Records API	Authenticated IPC interface used by all applications
SU-31.2	Data access layer	Transactions, concurrency, schema version checks
SU-31.3	Event data file store	Event data files referenced from the database
SU-31.4	Encryption at rest	Database and file encryption keys protected by the OS key store
SU-31.5	Integrity checker	Startup and scheduled database and file integrity checks
SI-32	Audit Trail and E-Signature Service	
SU-32.1	Audit capture	Receives audit events from the records API and firmware log imports
SU-32.2	Audit store	Append-only, hash-chained audit records
SU-32.3	Electronic signature service	Password re-entry, signature manifestation, record binding

6.5 Shared services decomposition

Unit	Name	Responsibility
SI-40	Security Service	
SU-40.1	Authentication	User ID and password verification, salted password hashes
SU-40.2	Roles and permissions	Role matrix (SRS-SW-CSI-010 Section 3.4)
SU-40.3	Session manager	Session tokens, idle lock, account lockout
SU-40.4	Security client	In-process client used by each application
SI-41	Instrument Communication Library	
SU-41.1	Connection and pairing	Mutual authentication, session arbitration, reconnect
SU-41.2	Command client	Command send, acknowledgement tracking, timeouts
SU-41.3	Telemetry subscriber	Status and alarm messages at 10 Hz
SU-41.4	Event stream receiver	High-rate event and image channel, buffering
SU-41.5	Protocol codec	Framing, sequence numbers, CRC-32, version check

7. Firmware Architecture (Part B)

7.1 Structure

Figure 4 shows the firmware items. SI-10 runs as fixed-priority tasks on a certified RTOS; the fault manager has the highest priority. SI-11 is a pipelined FPGA design with no software in the sort path. SI-15 runs on its own microcontroller and can remove the high-voltage and charge enables in hardware regardless of the state of SI-10 or SI-11.

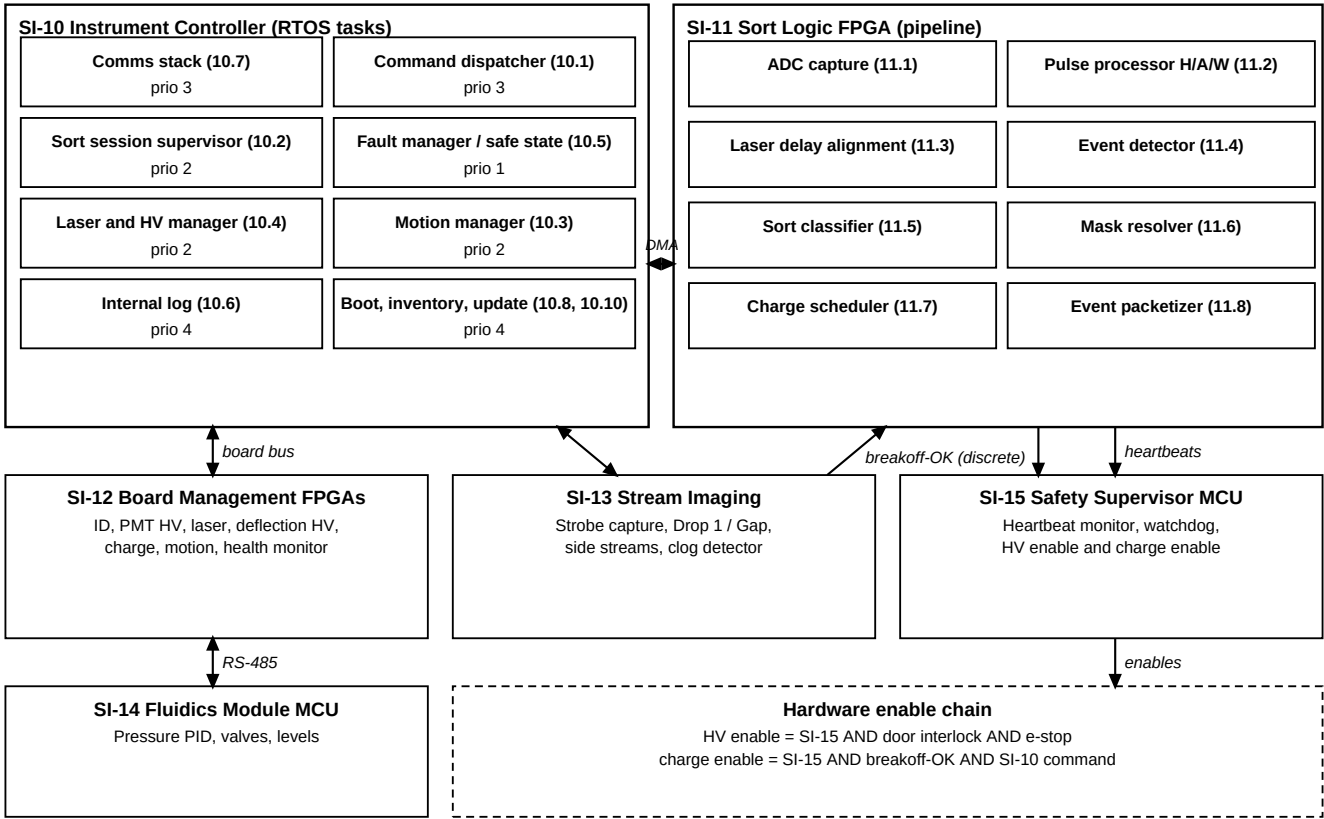


Figure 4. Firmware architecture.

7.2 Firmware decomposition

Unit	Name	Responsibility
SI-10	Instrument Controller Firmware	
SU-10.1	Command dispatcher	Validates and executes commands; state machine owner
SU-10.2	Sort session supervisor	Downloads gate tables and masks to SI-11; plate stage sequencing
SU-10.3	Motion manager	Loading port, aspirator drawer, filter and plate stage through SI-12
SU-10.4	Laser and HV manager	Laser on/off and power, PMT and plate voltage setpoints through SI-12
SU-10.5	Fault manager	Fault detection aggregation, safe state entry and latching (Section 10.4)
SU-10.6	Internal log	Non-volatile ring buffer, time stamps, export with checksum
SU-10.7	Comms stack	Instrument side of ICD-CSI-020: channels, CRC, sequence, heartbeat
SU-10.8	Boot and board inventory	Self test, FPGA CRC checks, board inventory
SU-10.9	Time and usage counters	Clock sync, laser/PMT/stream/sort/compressor hours
SU-10.10	Update manager	Signed update verification, manifest compatibility, rollback
SI-11	Sort Logic FPGA	
SU-11.1	ADC capture	Continuous sampling of all detector channels
SU-11.2	Pulse processor	Height, area, width per parameter
SU-11.3	Laser delay alignment	Per-laser delay and area scaling
SU-11.4	Event detector	Threshold, window extension, electronic abort counting
SU-11.5	Sort classifier	Gate lookup against downloaded tables
SU-11.6	Mask resolver	Yield, purity and phase masks at 1/32-drop resolution
SU-11.7	Charge scheduler	Drop clock, drop delay, charge timing; honors breakoff-OK and charge enable
SU-11.8	Event packetizer	Event records to DMA FIFO, sorted/not-sorted tags
SI-12	Board Management FPGA	
SU-12.1	Board identity and bus slave	Identity record, register map, bus interface
SU-12.2	PMT HV control	DAC setpoint and ADC readback per PMT
SU-12.3	Laser board control	Laser switching, power set, monitor readback, shared-spot lockout
SU-12.4	Deflection HV and charge control	Plate voltage and charge level setpoints and readback
SU-12.5	Motor driver control	Step/direction, position feedback, stall detection
SU-12.6	Health monitor	Rails, temperatures, fans
SI-13	Stream Imaging Firmware	
SU-13.1	Strobe and capture	Strobe phase, image capture from breakoff and side-stream cameras
SU-13.2	Breakoff analyzer	Drop 1 and Gap computation
SU-13.3	Side-stream analyzer	Stream positions, drop-delay bead counts
SU-13.4	Clog detector	Stream loss or deviation; breakoff-OK discrete output
SI-14	Fluidics Module Firmware	
SU-14.1	Pressure regulator	PID control of sheath pressure
SU-14.2	Valve and pump control	Compressor, vacuum, valves, venting
SU-14.3	Level sensing	Level probes with calibration
SU-14.4	Serial link	ICD-CSI-010 protocol

Unit	Name	Responsibility
SI-15	Safety Supervisor Firmware	
SU-15.1	Heartbeat monitor	Monitors SI-10 and SI-11 heartbeats (250 ms limit)
SU-15.2	Enable-line driver	Drives HV enable and charge enable outputs
SU-15.3	Watchdog	Independent hardware watchdog
SU-15.4	Fault report	Reports cause of enable removal to SI-10

7.3 Task model and timing (SI-10)

Task	Priority	Period / trigger	Worst-case budget
Fault manager (SU-10.5)	1 (highest)	Event-driven; 10 ms polling of fault inputs	1 ms
Laser and HV manager (SU-10.4)	2	100 ms and on command	2 ms
Sort session supervisor (SU-10.2)	2	On command; stage moves	5 ms
Motion manager (SU-10.3)	2	10 ms	1 ms
Comms stack (SU-10.7)	3	Message-driven; heartbeat 250 ms	3 ms per message
Command dispatcher (SU-10.1)	3	Message-driven	5 ms, acknowledgement within 100 ms
Internal log (SU-10.6)	4	Queue-driven; flush every 1 s	5 ms
Boot, update, usage (SU-10.8 to 10.10)	4	Startup, on command, 60 s	Not time-critical

Schedulability is confirmed by rate-monotonic analysis in SDS-FW-CSI-013 with at least 40% CPU margin at maximum event rate. The event data path uses DMA and does not consume task time proportional to event rate.

8. Interfaces

8.1 Internal interfaces

ID	Provider / consumer	Type	Data	Rate / timing	Safety
I-01	SI-03, SI-04, SI-05 / SI-41	In-process API	Commands, setups, gate tables	On demand	B
I-02	SI-41 / SI-10	Ethernet, TCP (ICD-CSI-020)	Command, telemetry, event, image, log channels	Events $\geq 110,000/s$; telemetry 10 Hz	B
I-03	All apps / SI-31	Local IPC, authenticated	Records, queries, audit events	On demand	B
I-04	All apps / SI-40	Local IPC via SI-31	Login, permissions, signatures	On demand	B
I-05	SI-10 / SI-11	On-chip bus, DMA FIFO	Gate tables, masks, setpoints; event records	Events at full rate	B
I-06	SI-10 / SI-12	Board bus, register map	Setpoints, readbacks, identity, health	≥ 10 Hz status	B
I-07	SI-10 / SI-13	High-speed serial	Drop 1, Gap, images, clog flag, strobe phase	≥ 10 frames/s	B
I-08	SI-13 / SI-11	Discrete signal	breakoff-OK	Updated every frame	B
I-09	SI-10 / SI-14	RS-485 (ICD-CSI-010)	Pressure setpoint, levels, faults	≥ 10 Hz	A
I-10	SI-10, SI-11 / SI-15	Discrete signals	Heartbeats; fault report	Heartbeat ≤ 100 ms	B
I-11	SI-15 / hardware	Discrete outputs	HV enable, charge enable	Removed within 10 ms of decision	B

8.2 Instrument protocol (summary of ICD-CSI-020)

The instrument link uses separate TCP connections for each channel so that high-rate event data cannot delay commands or telemetry.

Channel	Direction	Content	Priority
Command	Workstation to instrument, acknowledged	Commands with parameters; acknowledgement within 100 ms	Highest
Telemetry	Instrument to workstation	Status, alarms, fault and safe-state reports, 10 Hz; heartbeats every 250 ms both ways	High
Event	Instrument to workstation	Event records in blocks, 2 s instrument buffer	Bulk
Image	Instrument to workstation	Breakoff and side-stream images, at least 10 frames/s	Bulk
Log	Instrument to workstation, on request	Internal log export (service session only)	Low

Every message has a header (protocol version, channel, message type, sequence number, length), a payload and a CRC-32. Messages with a bad CRC, sequence error, unknown type or out-of-range parameter are rejected, logged and, for commands, acknowledged with a rejection reason. The connection is set up only after mutual authentication with the paired workstation's credentials.

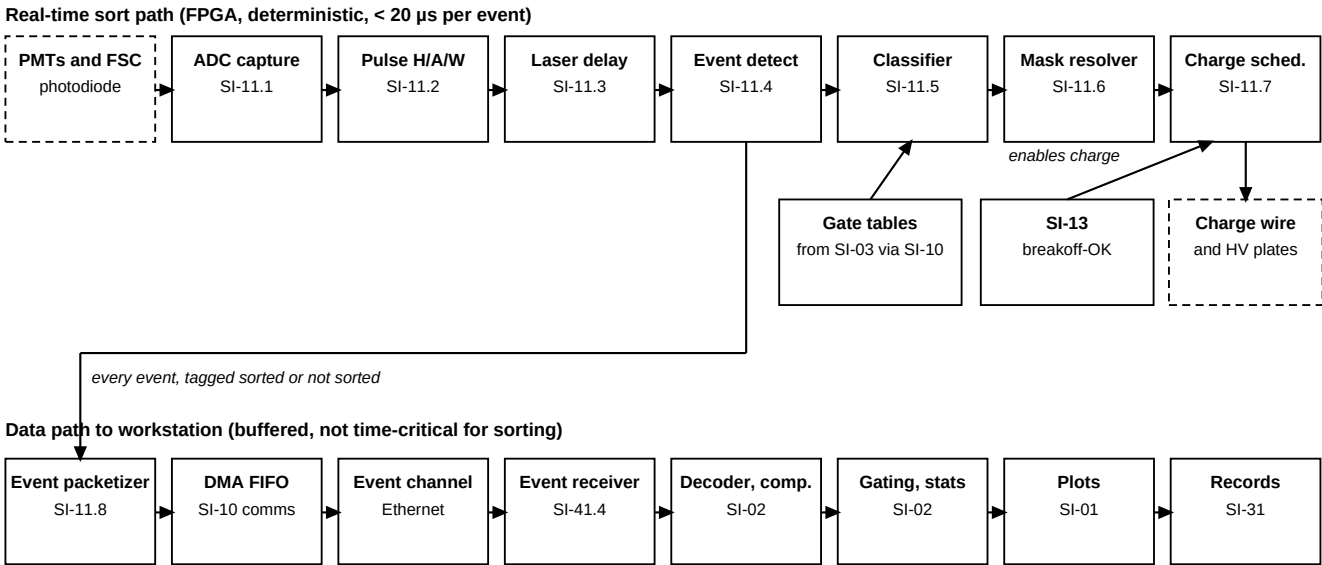
8.3 External interfaces

ID	Interface	Architecture element
IF-05	User interfaces of the three applications	Presentation layers (Figure 3)
IF-06	File export, LIS export, backup and archive	SU-30.3, SU-30.4
IF-07	Calibration reference readings (manual entry or USB import)	SU-21.1 to SU-21.4
Hardware	ADCs, DACs, laser control ports, motor drivers, cameras, level probes, interlock inputs	SI-11, SI-12, SI-13, SI-14, SI-15

9. Data Flows and Key Scenarios

9.1 Event data and sort decision flow

Figure 5 shows the two paths every event takes. The real-time path in the FPGA makes the sort decision and charges the drop; it never waits for the workstation. The data path carries every event, tagged as sorted or not sorted, to the Research Application for display, statistics and storage.



Sort decisions never depend on the workstation: gates, masks and drop delay are loaded into firmware before the sort starts.

Figure 5. Event data and sort decision flow.

9.2 Scenario: start a sort

Step	Element	Action
1	SU-03.5	Operator starts sort; checks stream, breakoff control, plate voltage, QC status and device (RA-701).
2	SU-03.4	Compiles gates and layout into gate tables; selects masks from the precision mode.
3	SU-41.2	Sends gate tables, masks, drop delay and target counts; waits for acknowledgement.
4	SU-10.2	Validates and loads tables into SI-11; enables charging if SI-15 charge enable and breakoff-OK are present.
5	SI-11	Classifies each event, resolves masks, schedules charge; tags and streams every event.
6	SU-03.5	Displays counters from telemetry; stops sort when targets are reached; SU-03.6 writes the sort report via SI-31.

9.3 Scenario: clog during sorting

Step	Element	Action
1	SU-13.4	Detects stream loss or Gap above limit within 500 ms; drops breakoff-OK and flags a clog to SI-10.
2	SU-11.7	Stops charging immediately when breakoff-OK drops.
3	SU-10.5	Enters the safe state: stream off, plate voltage off, drawer closed, tube unloaded, within 2 s; latches the fault.
4	SU-10.6	Logs the event with cause and time.
5	SU-41.3 / SU-01.4	Receives the safe-state report; shows the high clog alarm and recovery procedure (RA-504).
6	SU-10.5	Leaves the safe state only after cause cleared and authenticated acknowledgement (FW-607).

9.4 Scenario: loss of workstation link

Step	Element	Action
1	SU-10.7	Misses heartbeats for 2 s.
2	SU-10.5	During a sort: stops charging and turns off plate voltage, keeps the stream running (FW-603); logs the event.
3	SU-15.1	Independently, if SI-10 itself stops sending heartbeats for 250 ms, removes HV and charge enables in hardware.
4	SU-41.1	On reconnect: re-authenticates, checks protocol version, reads the fault state and shows it to the operator.

9.5 Scenario: daily performance check and IVD lockout

Step	Element	Action
1	SU-05.1	Loads the released baseline and bead lot for the active configuration from SI-31.
2	SU-05.1	Runs QC beads; computes robust CV, intensity and voltage per detector; sets laser delay and area scaling.
3	SU-05.2	Evaluates pass/warn/fail and stores the result; starts the 24-hour validity period.
4	SU-05.3	Before running an IVD template, checks QC validity, configuration match (SU-05.4) and calibration status; blocks if any fails.

9.6 Scenario: hardware configuration change and calibration

Step	Element	Action
1	SU-41.1	Admin Application requests an exclusive service session; firmware grants it when idle.
2	SU-20.2	Reads board inventory from firmware and shows differences from the configuration record.
3	SU-20.1 / SU-20.3	Service engineer edits the record; rules engine checks CFG-CSI-001 rules.
4	SU-32.3	Change is signed; SI-31 stores a new configuration version; SU-20.4 generates the optical configuration.
5	SU-21.x	Required calibrations run; SU-21.5 stores as-found and as-left values.
6	SU-22.5	New baseline is created; IVD templates are unblocked only after it is released.

9.7 Scenario: IVD result review and release

Step	Element	Action
1	SU-05.3	IVD run completes; result stored as Draft with template version, QC status and flags.
2	SU-30.2	Supervisor reviews result in the Data Management Application and marks it Reviewed.
3	SU-32.3	Supervisor signs release; signature bound to the record; audit entry created by SU-32.1.
4	SU-30.3	Released result can be printed or exported to the LIS; export carries a checksum.

10. Safety Architecture and Segregation

10.1 Risk control allocation

Table 2 shows how each hazard that software can contribute to is controlled in the architecture. Hardware controls are listed first because they do not depend on software.

Hazard	Hardware control	Software architecture control	Items
HA-01 Laser exposure	Enclosure, shutter interlock on flow cell door	Firmware monitors interlock state and rejects conflicting commands	SI-10
HA-02 High voltage	Door interlock and e-stop remove HV; enable chain	SI-15 removes HV enable on missed heartbeat; fault manager safe state; readback checks	SI-10, SI-12, SI-15
HA-04 Aerosol after clog	Sealed doors, evacuation, ULPA filter	Clog detection in SI-13; autonomous firmware response; recovery guidance	SI-13, SI-10, SI-01
HA-06 Moving parts	Hardwired e-stop	Firmware stops motion; stall detection	SI-10, SI-12
HA-07 Breakoff drift, low purity	—	breakoff-OK discrete line gates charging in FPGA within 100 ms	SI-13, SI-11
HA-08 Wrong result from configuration or QC	—	Board inventory check, configuration rules, QC lockout, assay lock, calibration due check	SI-10, SI-20, SI-05, SI-21
HA-09 Overpressure	Relief valves	Pressure regulation and venting in SI-14	SI-14
HA-12 Electrical fault	Circuit breaker, insulation	Rail and temperature monitoring leading to safe state	SI-12, SI-10
HA-15 Fluid run-out during sort	—	Level monitoring and capacity check before sort	SI-14, SI-04

Table 2. Risk control allocation (hazards from HA-CSI-001).

10.2 Segregation

Segregation	Mechanism	Purpose
Safety supervisor from controller	Separate MCU, own clock and watchdog, hardware enable outputs	HV and charge are removed even if SI-10 or SI-11 fails
Sort path from software	SI-11 is FPGA logic; no CPU in the sort decision path	Timing and correctness independent of RTOS load
Class A fluidics firmware	Separate MCU in the fluidics module; only the ICD-CSI-010 serial link to SI-10	A failure of SI-14 cannot corrupt Class B items
Applications from each other	Separate Windows processes; separate permissions	A fault in one application cannot corrupt another
Data from applications	Records service is the only database writer; authenticated IPC	Protects data integrity and the audit trail
Workstation from instrument	Firmware validates every command; safety responses local to firmware	Workstation faults cannot create unsafe instrument states

10.3 Instrument state model

State	Description	Entered from / by	Allowed outputs
Boot	Self test, FPGA CRC, board inventory	Power-on	None
Safe	HV off, charging off, motion stopped, drawer closed	Boot; any fault; e-stop; clog	Lasers as set; stream per cause
Idle	Ready for commands, no stream	Safe after acknowledgement	Lasers, fluidics procedures
Stream	Stream on, breakoff control available	Idle by command	Stream, drop drive, cameras
Acquire	Sample loaded, events streamed	Stream by Load	Plus sample pressure
Sort	Charging enabled, plate voltage on	Acquire by sort start (RA-701 checks)	Plus HV and charging
Paused	Charging off, acquisition continues	Sort on breakoff out of range or user pause	As Acquire
Service	Exclusive Admin Application session	Idle by service request	Calibration functions

Any fault moves the instrument to Safe from any state. Safe is latched until the cause clears and an authenticated user acknowledges it (FW-607).

10.4 Fault manager design

SU-10.5 collects fault inputs from SI-11 (overflow, charge timing), SI-12 (readback, rails, temperature), SI-13 (clog), SI-14 (pressure), SI-15 (fault report), the interlock inputs and the comms stack. Each fault has a defined reaction (safe state with stream off, safe state with stream on, or warning only) in a table reviewed as part of risk management. The fault manager runs at the highest task priority and its reactions do not require any workstation message.

11. Data Architecture

Store	Owner	Content	Protection
Records database	SU-31.2	Experiments, results, configurations, setups, calibrations, baselines, users, signatures	Transactions, encryption at rest, integrity checks
Event data files	SU-31.3	Event data referenced by records	Checksums, encryption at rest
Audit store	SU-32.2	Audit entries from all applications and security events	Append-only, hash-chained, not editable
Internal log	SU-10.6	Firmware events, faults, usage counters	Non-volatile ring buffer; export with checksum
Board identity records	SU-12.1	Part number, serial, revision, calibration data	Written in manufacturing; read-only in field
Fluidics calibration	SI-14	Pressure and level sensor calibration	Written only in a service session

Time: the workstation is the time source for the instrument (FW-808); all records and audit entries use UTC with the workstation's configured time zone for display.

12. Security Architecture

Trust boundary	Threat	Control
Operator to application	Unauthorized use or change	SI-40 authentication, roles, idle lock, account lockout
Application to records service	Tampering with records	Authenticated IPC; only SI-31 writes; audit trail
Workstation to instrument	Spoofed or corrupted commands	Pairing and mutual authentication; CRC and sequence checks; range checks in firmware
Update path	Malicious software or firmware	Signed packages and manifest; verification before install; rollback
Workstation to network	Network attack	Locked image, minimal services, isolated instrument link, security updates validated before release
Exported data	Undetected alteration, privacy	Checksums; de-identified export option

The threat model and cybersecurity risk assessment are in the cybersecurity management file; the SBOM is generated for every release (CM-203).

13. SOUP

Per IEC 62304 clauses 5.3.3 and 5.3.4, Table 3 gives the functional and performance requirements of each SOUP item and the hardware and software it needs. Published anomaly lists are reviewed for each version before release and the evaluation is recorded in the SOUP register.

SOUP (version)	Used by	Required function and performance	Hardware / software needed
Windows 10 IoT Enterprise LTSC	All workstation items	Supported for product life; locked configuration	Validated PC, 32 GB RAM, SSD
.NET runtime (LTS)	Workstation items	Application runtime; LTS support	Windows as above
Embedded SQL database engine	SI-31	ACID transactions; encryption support	.NET; local SSD
Plotting library	SI-01, SI-30	1,000,000 events per plot at ≥ 2 updates/s	GPU with DirectX 11
Cryptographic library	SI-40, SI-31, SI-41, SI-10	FIPS 140-validated hashing, signatures and TLS	Workstation and controller cores
Certified RTOS	SI-10	Deterministic fixed-priority scheduling	Controller SoC processor cores
FPGA vendor IP cores	SI-11, SI-12	ADC interface, DSP, bus; timing closure at design clock	Controller SoC logic, board FPGAs
Embedded Linux and image library	SI-13	Process ≥ 10 frames per second	Imaging processor

Table 3. SOUP items.

14. Architectural Design Decisions

ID	Decision	Alternative considered	Rationale	Consequences
ADR-01	Sort decisions in FPGA logic	Sort decision in RTOS software on the controller CPU	Deterministic timing under 20 μ s at full event rate; no dependence on CPU load	FPGA verification by simulation and HIL; changes need FPGA builds
ADR-02	Autonomous safety response in firmware	Workstation software commands the safe state	Safe state reached within 2 s even if the workstation fails	Fault reaction table must be kept consistent between firmware and applications
ADR-03	Independent safety supervisor MCU	Rely on SI-10 watchdog only	HV and charge removed even if SI-10 or SI-11 hangs; supports Class B rationale	Additional part and firmware item to verify
ADR-04	Three applications on shared services	One application with role-based menus	Clear separation of routine use, calibration and records; smaller verification scope per change	Shared services must be versioned together
ADR-05	Records service as the only database writer	Each application writes directly to the database	One place for transactions, audit capture and encryption; Part 11 compliance	IPC interface on every data path
ADR-06	Channelized TCP protocol with CRC and heartbeats	Single UDP stream	Reliable delivery for commands; bulk events cannot block commands; fast link-loss detection	Slightly higher latency than UDP for events (buffered)
ADR-07	Configuration-driven optics and setups	Hard-coded configurations per build	16 builds and field upgrades without software changes; controlled records with signatures	Rules engine must be verified for every rule in CFG-CSI-001
ADR-08	breakoff-OK as a discrete signal	Breakoff status sent as a software message	Charging stops within 100 ms regardless of software load	Extra hardware line between imaging processor and FPGA
ADR-09	Signed updates and locked workstation image	Customer-managed updates	Cybersecurity and validated state preserved	Manufacturer must deliver validated OS updates

15. Integration Strategy

15.1 Approach

Integration is incremental and bottom-up, following the dependencies in Figure 4 and Figure 3. Firmware is integrated with hardware first, then the instrument protocol, then the applications one part at a time. Each increment has entry and exit criteria and is regression-tested with all earlier increments. Integration tests are specified in ITS-CSI-010.

Increment	Content	Environment	Exit criteria
INC-1 FPGA pipeline	SI-11 units in simulation with recorded detector data	FPGA simulation	Sort decisions match reference model bit-exactly; timing closure
INC-2 Board bring-up	SI-12 on each board type; SI-10 board bus and inventory	Board test fixtures	All registers, setpoints and readbacks verified; identity records read
INC-3 Controller and safety	SI-10 with SI-11, SI-12, SI-15 on the controller; fault manager and state model	HIL bench with simulated interlocks	All fault reactions and safe state timing verified
INC-4 Imaging and fluidics	SI-13 with cameras and SI-11 breakoff-OK; SI-14 with fluidics module	Integration instrument	Clog detection, breakoff pause, pressure control verified
INC-5 Instrument protocol	SI-41 with SI-10 comms stack; all channels	HIL bench and network emulator	Throughput $\geq 110,000$ events/s; CRC, sequence, heartbeat and reconnect behavior
INC-6 Records and security	SI-31, SI-32, SI-40 with a test client	Workstation	Transactions, audit chain, signatures, encryption, recovery after crash
INC-7 Research Application	SI-01 to SI-05 with instrument and records service	Integration instrument	All Part A integration tests pass
INC-8 Admin Application	SI-20 to SI-22 with instrument service session	Integration instrument	Configuration control and all calibrations verified
INC-9 Data Management Application	SI-30 with records from INC-7 and INC-8	Workstation	Review, release, export, backup and archive verified
INC-10 Full system	All items on production-equivalent instruments, all configurations	System test lab	Ready for software system test (STS-CSI-010)

15.2 Integration across subsystems

Software integration is coordinated with hardware subsystem integration in the system integration plan. Firmware increments INC-2 to INC-4 are scheduled against board and subsystem availability (SS-01 to SS-09). Simulators replace missing hardware: a detector signal generator for SI-11, an interlock and enable-line simulator for SI-15, and an instrument simulator implementing ICD-CSI-020 for application development before INC-5.

15.3 Continuous integration and regression

- Every commit builds all workstation items and firmware images and runs unit tests (UTS-CSI-010 series) and static analysis.
- Nightly builds run FPGA simulation regression and the HIL integration suite for firmware.
- Release candidates run the full integration suite (ITS-CSI-010) and the regression part of STS-CSI-010.
- Any failed safety-related test blocks the build from promotion.

16. Requirements Allocation and Architecture Verification

16.1 Allocation of software requirements to items

Generated from the Item column of SRS-SW-CSI-010 Rev E. Every software requirement is allocated to at least one item defined in this architecture; unit-level allocation is in SDS-SW-CSI-012 and SDS-FW-CSI-013.

Item	Units	Allocated software requirements
SI-01	5	RA-101, RA-102, RA-103, RA-104, RA-105, RA-504, RA-702, RA-1001, RA-1002, RA-1003, CM-301, CM-302, CM-305
SI-02	5	RA-301, RA-302, RA-303, RA-304, RA-305, RA-306, RA-307, RA-901, RA-903, CM-303
SI-03	6	RA-401, RA-402, RA-403, RA-404, RA-405, RA-406, RA-407, RA-501, RA-502, RA-503, RA-505, RA-601, RA-602, RA-603, RA-604, RA-605, RA-701, RA-702, RA-703, RA-704, RA-705, RA-706
SI-04	4	RA-201, RA-202, RA-203, RA-204, RA-205, RA-206, RA-207, RA-208, RA-1002
SI-05	4	RA-701, RA-801, RA-802, RA-803, RA-804, RA-805, RA-806, RA-901, RA-902, RA-903, RA-904, AD-104, AD-203, AD-903
SI-10	10	RA-204, RA-205, RA-505, RA-805, FW-101, FW-103, FW-104, FW-208, FW-302, FW-402, FW-501, FW-502, FW-601, FW-602, FW-603, FW-605, FW-606, FW-607, FW-608, FW-701, FW-702, FW-703, FW-704, FW-705, FW-706, FW-801, FW-802, FW-803, FW-804, FW-805, FW-806, FW-807, FW-808, AD-102, CM-203, CM-303
SI-11	8	RA-802, FW-102, FW-201, FW-202, FW-203, FW-204, FW-205, FW-206, FW-207, FW-806
SI-12	6	FW-102, FW-301, FW-302, FW-303, FW-304, FW-305, FW-306, FW-307, FW-308, FW-309, FW-310, FW-606, AD-502, AD-504
SI-13	4	RA-404, RA-407, RA-501, FW-205, FW-209, FW-210, AD-501
SI-14	4	FW-401, FW-402, FW-403, AD-401, AD-403
SI-15	4	FW-103, FW-604, FW-605
SI-20	4	AD-101, AD-102, AD-103, AD-104, AD-105, AD-106, AD-107, AD-305
SI-21	6	AD-201, AD-202, AD-203, AD-204, AD-205, AD-301, AD-302, AD-303, AD-304, AD-305, AD-401, AD-402, AD-403, AD-404, AD-405, AD-501, AD-502, AD-503, AD-504, AD-505, AD-601, AD-901, AD-902, AD-903, AD-904, AD-905
SI-22	5	AD-601, AD-602, AD-603, AD-701, AD-702, AD-703, AD-801, AD-802
SI-30	5	DM-101, DM-102, DM-105, DM-203, DM-301, DM-304, DM-401, DM-402, DM-403, DM-404, DM-501, DM-502, DM-503, CM-202, CM-203, CM-304
SI-31	5	RA-307, RA-705, RA-706, RA-904, AD-902, DM-103, DM-104, DM-105, DM-504
SI-32	3	AD-103, AD-701, AD-802, DM-201, DM-202, DM-204, DM-302, DM-303, DM-304, DM-503
SI-40	4	RA-804, AD-801, DM-204, CM-101, CM-102, CM-103
SI-41	5	RA-303, FW-801, FW-802, FW-805, FW-806, FW-807, FW-808, CM-201, CM-302

Check: 179 software requirements in SRS-SW-CSI-010; 0 unallocated; 19 items in this architecture, 0 items referenced by requirements but not defined here.

16.2 Architecture verification (IEC 62304 5.3.6)

Criterion	How verified	Result
Architecture implements all software requirements	Allocation table 16.1 generated from SRS-SW-CSI-010	All requirements allocated
Interfaces between items and to SOUP are defined	Review of Section 8 against ICD-CSI-010, ICD-CSI-020 and SDS register map	Complete
Architecture supports risk controls	Review of Table 2 against HA-CSI-001 and RMR-CSI-001	All software-related hazards allocated
Segregation is effective	Review of Section 10.2; fault injection tests in INC-3	Planned and executed in ITS-CSI-010
Timing requirements are achievable	Rate-monotonic analysis; FPGA timing closure	Margins recorded in SDS-FW-CSI-013
SOUP requirements are specified	Review of Section 13 and SOUP register	Complete

The architecture review was held with software, firmware, systems, quality and cybersecurity reviewers; minutes and action items are filed in the design history file.

17. Reference Documents

17.1 External standards

Reference	Title
IEC 62304:2006 + A1:2015	Medical device software: software life cycle processes
ISO 14971:2019	Medical devices: application of risk management to medical devices
IEC 82304-1	Health software: general requirements for product safety
IEC 81001-5-1	Health software and health IT systems: security activities in the product life cycle
21 CFR Part 11	Electronic records; electronic signatures
FDA premarket cybersecurity guidance	Cybersecurity in medical devices

17.2 Program documents

Document no.	Rev	Title
SRS-CSI-001	D	System Requirements Specification
SRS-SW-CSI-010	E	Software Requirements Specification: Research, Firmware, Admin and Data Management
SAD-CSI-001	C	System Architecture Description
SDS-SW-CSI-012	B	Application Detailed Design Specification and User Interface Style Guide
SDS-FW-CSI-013	B	Firmware and FPGA Detailed Design Specification (including board register map)
ICD-CSI-010	B	Interface Control Document: Sorter to Fluidics Supply Module
ICD-CSI-020	A	Interface Control Document: Sorter to Workstation
CFG-CSI-001	C	Configuration Definition: Laser and Detector Builds
SDP-CSI-010	C	Software Development Plan
SCMP-CSI-010	B	Software Configuration Management Plan
SVP-CSI-010	B	Software Verification Plan
ITS-CSI-010	A	Software Integration Test Specification
STS-CSI-010	A	Software System Test Specification
HA-CSI-001	C	Hazard Analysis
RMR-CSI-001	A	Risk Management Report

Appendix A. Terms and Abbreviations

Term	Meaning
ADR	Architectural design decision record
breakoff-OK	Discrete signal from SI-13 that is true while Drop 1 and Gap are within limits
DMA	Direct memory access
FPGA / CPLD	Field-programmable gate array / complex programmable logic device
HIL	Hardware-in-the-loop test bench
IPC	Inter-process communication
RTOS	Real-time operating system
Safe state	Instrument state defined in Section 10.3
SI-nn / SU-nn.m	Software item / software unit identifiers
SoC	System on chip (processor cores and programmable logic)
SOUP	Software of unknown provenance